



2025

U23(B)

BOOKLET NO.

231096

Mechanical Engineering
Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional / Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

Q1. Solve the following : **(5×10=50)**

Q1. (a) Define entropy. What are the important causes of irreversibility ? **10**

Q1. (b) Explain how fluid flows are classified. **10**

Q1. (c) Explain development of thermal boundary layer over a hot flat plate with a neat sketch. **10**

Q1. (d) Derive general three dimensional heat conduction equation in Cartesian coordinates. **10**

Q1. (e) In a double pipe heat exchanger, hot water flows at the rate of 5000 kg/hr and gets cooled from 95°C to 65°C. At the same time 50000 kg/hr of cooling water enters into heat exchanger at 30°C. The flow conditions are such that the overall heat transfer coefficient remains constant at 2270 W/m².K for parallel flow condition.

Determine :

(i) Heat transfer area

(ii) Effectiveness of heat exchanger.

Take specific heat of water as 4.18 kJ/kg. K. **10**

Q2. (a) What is available energy and unavailable energy ? Derive an expression for decrease in available energy in a heat transfer process when it takes place through a finite temperature difference. **20**

Q2. (b) Define (i) Number of Transfer Units (NTU), and (ii) (Capacity Ratio CC). For parallel flow heat exchanger, prove that effectiveness ϵ is given by

$$\epsilon = \frac{1 - e^{-NTU(1+C)}}{1+C} \quad \text{10}$$

Q2. (c) Explain working of closed cycle gas turbine plant with the help of pressure-volume diagram. Write an expression for thermal efficiency. **10**



Q3. (a) Air at 200 kPa and 900 K enters an adiabatic nozzle at low velocity and is discharged at a pressure of 80 kPa. If the isentropic efficiency of the nozzle is 90%, determine

- (i) the maximum possible exit velocity,
- (ii) the exit temperature, and
- (iii) the actual exit velocity.

Represent the process on temperature-entropy diagram.

Take specific heat of air at constant pressure $C_p = 1.01$ kJ/kg and ratio of specific heats $\gamma = 1.35$.

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Q3. (b) Air at 20°C and 1 atmosphere flows over a plate at 50 m/s. The plate is 80 cm. long and 1 m wide and is maintained at 60°C. Determine heat transfer rate from the plate.

Use following properties of air at 40°C.

Prandtl No. $Pr = 0.7$

Conductivity $K = 0.02723$ W/m.K

$C_p = 1.007$ kJ/kg.K

Dynamic viscosity $\mu = 1.906 \times 10^{-5}$ kg/m.s

Use following correlations for Nusselt number N_u .

$$N_u = (0.036 Re_L^{0.8} - 871) \times Pr^{(1/3)} \text{ for } Re_L > 5 \times 10^5$$

$$N_u = 0.664 Re_L^{(1/2)} Pr^{(1/3)} \text{ for } Re_L < 5 \times 10^5.$$

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Q3. (c) Explain working of centrifugal compressor with a neat sketch.

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P.T.O.



- Q4. (a)** Explain Clausius and Kelvin Planck's statements of second law of thermodynamics. Prove that these two statements are equivalent in all respects. **20**
- Q4. (b)** A furnace wall lining is made up of a material having conductivity $2 \text{ W/m} \cdot ^\circ\text{C}$. Temperatures of the inner and outer surfaces of the lining are 800°C and 300°C respectively. Outer surface of the lining is exposed to ambient air at 20°C with convective heat transfer coefficient of $10 \text{ W/m}^2 \cdot \text{K}$.
Calculate :
(i) Heat flux,
(ii) Thickness of lining, and
(iii) Thickness of lining required if heat flow is to be reduced by 40%.
Draw labelled sketches. **20**
- Q4. (c)** Explain following basic laws of radiation.
(i) Stefan Boltzman Law
(ii) Wein's Displacement Law. **10**
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SECTION – B

- Q5.** Solve the following : **(5×10=50)**
- Q5. (a)** Explain working of four stroke petrol engine cycle with the help of pressure- volume diagram. **10**
- Q5. (b)** What is co-generation ? In which industries co-generation is used ? **10**
- Q5. (c)** Explain vapour compression refrigeration cycle with the help of pressure-enthalpy diagram. **10**
- Q5. (d)** Suggest alternate fuels for use in internal combustion engines. State advantages and disadvantages of any two alternate fuels. **10**
- Q5. (e)** Explain various methods used for heat recovery in steam power plants. **10**
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Q6. (a) How internal combustion engine fuels are rated ?

An automobile industry producing heavy duty trucks uses internal combustion engines as power plants for trucks. Elaborate on the procedure used for rating fuels used in these engines.

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Q6. (b) Determine thermal efficiency, heat supplied in boiler and turbine work of a Carnot cycle if boiler pressure is 180 bar and condenser pressure is 0.2 bar. Represent the cycle on temperature-entropy diagram. Use following table for steam properties.

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Saturation properties for steam-pressure table							
Pressure (MPa)	Temperature (°C)	Volume (m ³ /kg)		Enthalpy (kJ/kg)		Entropy (kJ/kg.K)	
		v_f	v_g	h_f	h_g	s_f	s_g
0.002	17.5	0.001	66.99	73.4	2544.8	0.3543	8.5764
0.02	60.06	0.00102	7.648	251.4	2624.5	0.9441	7.7675
0.2	120.21	0.00106	0.8857	504.7	2724.9	1.6717	6.9916
1.8	207.11	0.00117	0.1104	884.5	2795.9	2.3975	6.3775
2.0	212.38	0.00118	0.0996	908.5	2798.3	2.4468	6.339
18.0	356.99	0.00184	0.0075	1732.1	2509.8	3.8718	5.1061
20.0	365.75	0.00204	0.0059	1827.2	2412.3	4.0156	4.9314

Q6. (c) Represent following psychrometric processes on a hand drawn psychrometric chart.

- (i) Sensible heating
- (ii) Sensible cooling
- (iii) Humidification and
- (iv) Dehumidification.

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P.T.O.



- Q7. (a)** What is the necessity of compounding in steam turbines ? What are the different compounding methods used ? Explain any one method used for compounding in steam turbines. 20

- Q7. (b)** A vapour compression cycle refrigerator works between pressure limits of 530 kPa and 210 kPa. The vapour is superheated at the end of compression and its temperature is 310 K. The vapour is superheated by 10 degrees before entering into the compressor. The specific heat at constant pressure of superheated vapour is 0.6 kJ/kg. K.

Represent the cycle on temperature-entropy and pressure-enthalpy planes. Use data given below and find Coefficient Of Performance (C.O.P.) of the refrigerator. 20

Pressure (bar)	Saturation Temperature (°C)	h_f (kJ/kg)	h_{fg} (kJ/kg)
5.3	15.5	56.15	144.9
2.1	- 14.0	25.12	158.7

- Q7. (c)** Explain working principle of convergent-divergent nozzle used for expansion of steam with a neat sketch. 10

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- Q8. (a)** A full load test on a two stroke single cylinder engine yielded following observations.

Engine speed = 450 rpm

Brake load = 450 N

Brake drum diameter = 1.5 m

Rise in temperature of cooling water = 36°C

Cooling water flow rate = 440 kg/hr

Mass of fuel burnt = 5.4 kg/hr



Mass of exhaust gases discharged = 172.8 kg/hr

Temperature of exhaust gases = 355°C

Temperature of laboratory = 20°C

Draw up a heat balance sheet on percentage basis

Calorific value of fuel is 44000 kJ/kg.

The engine cylinder diameter is 22 cm and piston stroke is 27 cm.

Take specific heat for water as 4.18 kJ/kg.K and for exhaust gases as 1.005 kJ/kg.K.

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- Q8. (b) Explain ideal reheat Rankine cycle with the help of a neat sketch and temperature-entropy diagram. How it improves thermal efficiency as compared with a Rankine cycle ?

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- Q8. (c) What is boiler draught ?
Compare : Natural draught and Artificial draught.

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